AN ENGINEERING DECISION

A queue is not extra capacity.

Buffer the burst. Budget the recovery.

01

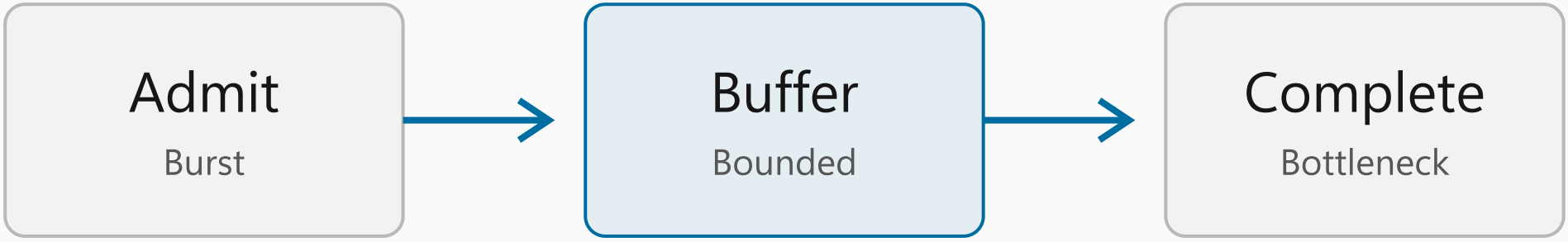
Capacity and waiting

THE ILLUSION

Fast acceptance.
Unchanged capacity.

The wait moved. The work did not.

A waiting room.
Not a wider exit.



RECOVERY

Only spare capacity drains the backlog.

- Completions must outrun new arrivals.
- Full utilization leaves nothing for recovery.

02

The hidden cost of acceptance

Watch the oldest unfinished promise.

- Queue depth counts work.
- Job age exposes delay.
- Include in-flight jobs.

Retries are more arrivals.

- One identity per logical job.
- One finite retry budget.

03

A bounded promise

ADMISSION

Bound the promise, not just the queue.

- Useful completion deadline.
- Hard storage limit.
- Honest rejection.

Every job needs a visible ending.

- Completed.
- Failed.
- Explicitly expired.

THE DECISION TEST

Prove recovery under continuing load.

> Burst. Slow dependency. Worker restart.

> Account for every accepted job.

THE KEY MESSAGE

Buffer the burst.
Budget the recovery.

A queue buys time, not capacity.